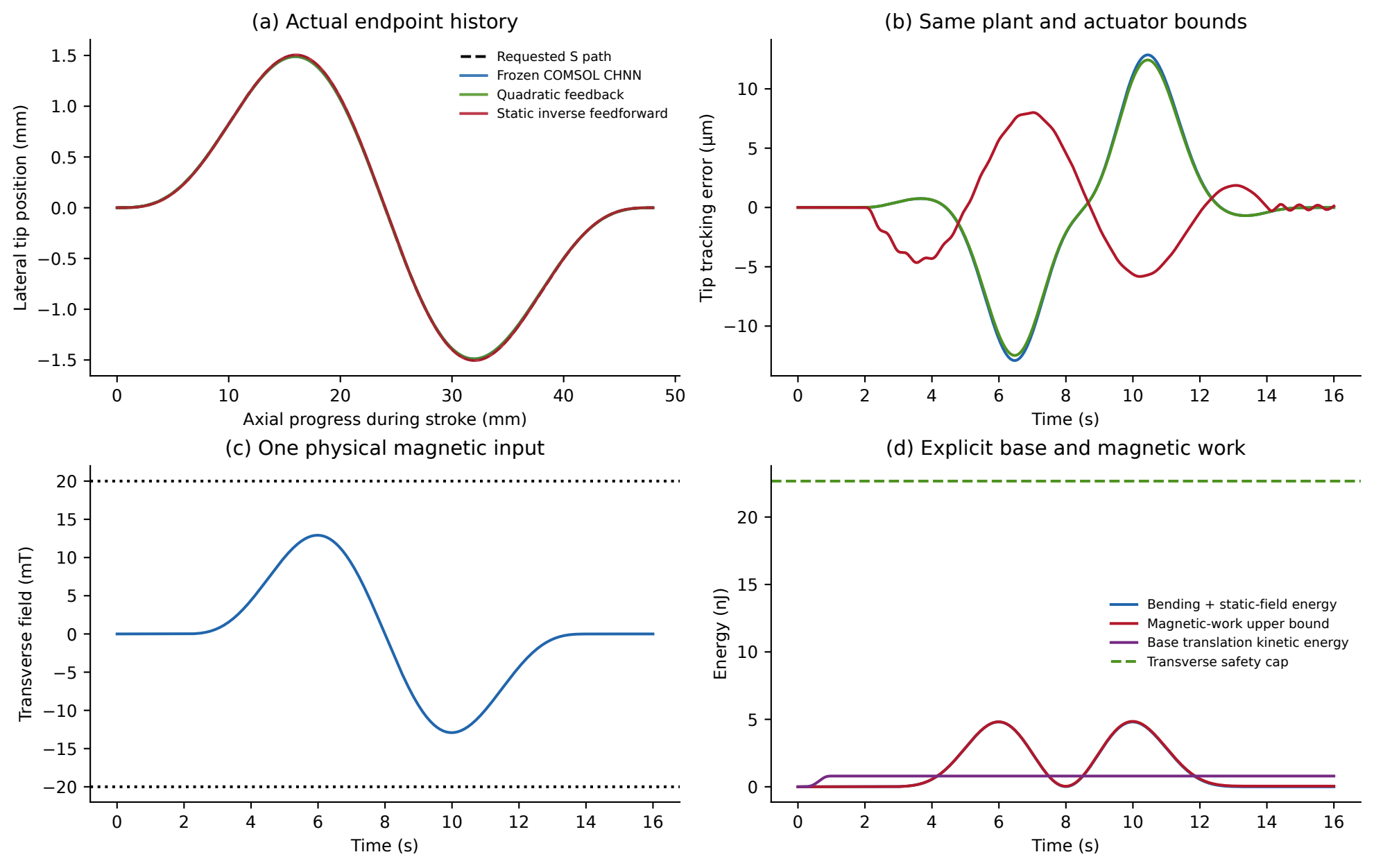
(a) Actual endpoint history
Requested S path
Frozen COMSOL CHNN
Quadratic feedback
Static inverse feedforward
Lateral tip position (mm)
Axial progress during stroke (mm)
(b) Same plant and actuator bounds
Tip tracking error (μm)
Time (s)
(c) One physical magnetic input
Transverse field (mT)
Time (s)
(d) Explicit base and magnetic work
Bending + static-field energy
Magnetic-work upper bound
Base translation kinetic energy
Transverse safety cap
Energy (nJ)
Time (s)